Order-Retract Monotonicity for SEP Data Hiding

Candidate partial result packet for arXiv:1703.03392

Run `fa_banach_001`; prepared 2026-07-01

Source Problem

The source paper is Ludovico Lami, Carlos Palazuelos, and Andreas Winter, *Ultimate data hiding in quantum mechanics and beyond*, arXiv:1703.03392. In the conclusion the authors ask about the opposite end of the separable-measurement data hiding ratio. They observe that if one local cone is simplicial, then no data hiding against separable measurements is possible, and they state the converse as a natural conjecture:

$$R(\mathrm{SEP}) = 1 \quad \text{if and only if} \quad C_A \text{ or } C_B \text{ is simplicial.}$$

The same paragraph records the known Namioka–Phelps case: if C_B is the cone of the gbit, then $R(\mathrm{SEP}) = 1$ if and only if C_A is simplicial.

> Lastly, while here we have focused on the maximum data hiding ratios among $R(\mathrm{SEP})$ for various local GPTs, the opposite end is very interesting, too, from a foundational point of view. Namely, it is clear that to have $R(\mathrm{SEP}) > 1$, it is necessary that both cones C_A and C_B are non-classical in the sense that they can not be simplicial, for the reason that otherwise min- and max-tensor product of the cones would coincide. It is however not known that this is necessary, and one might conjecture that $R(\mathrm{SEP}) = 1$ if and only if one of the cones C_A or C_B is simplicial. A general result along these lines is due to Namioka and Phelps [27], which says that for C_B the cone of the so-called gbit, $R(\mathrm{SEP}) = 1$ iff C_A is simplicial. If the more general conjecture were true, it would make a good case for $R(\mathrm{SEP})$ as a measure of non-classicality of a GPT. For a thorough discussion of this and related problems, see [1, Chapter II].

Definitions

All GPTs below are finite-dimensional and use the source paper's convention: a GPT is a triple $A = (V_A, C_A, u_A)$ with base norm

$$\|x\|_A = \min\{u_A(x_+) + u_A(x_-) : x_\pm \in C_A,\ x = x_+ - x_-\}.$$

For two local GPTs A, B, $A \otimes_{\min} B$ denotes the minimal composite, i.e. the cone generated by $C_A \otimes C_B$. The ratio

$$R_{A\otimes_{\min}B}(\mathrm{SEP}) = \max_{0\neq X\in V_A\otimes V_B} \frac{\|X\|_{A\otimes_{\min}B}}{\|X\|_{\mathrm{SEP}}}$$

is the source paper's formula for the data hiding ratio against separable measurements, specialized to the minimal composite.

Say that $A_0 = (V_0, C_0, u_0)$ is a *local order-retract subtheory* of $A = (V, C, u)$ if there are linear maps

$$i : V_0 \to V, \qquad p : V \to V_0$$

such that i and p are positive, preserve the order units

$$u \circ i = u_0, \qquad u_0 \circ p = u,$$

and satisfy $pi = \mathrm{id}_{V_0}$.

Claimed Partial Result

Theorem 1 (Order-retract monotonicity)**.** *Let A_0 and B_0 be local order-retract subtheories of A and B. Then*

$$R_{A\otimes_{\min}B}(\mathrm{SEP}) \geq R_{A_0\otimes_{\min}B_0}(\mathrm{SEP}).$$

More precisely, if $I = i_A \otimes i_B$, then for every $X \in V_{A_0} \otimes V_{B_0}$,

$$\|IX\|_{A\otimes_{\min}B} = \|X\|_{A_0\otimes_{\min}B_0}, \qquad \|IX\|_{\mathrm{SEP}(A,B)} = \|X\|_{\mathrm{SEP}(A_0,B_0)}.$$

Proof Intuition

A positive unit-preserving map between GPTs is a contraction for the base norm, because any positive decomposition of x is sent to a positive decomposition with the same total unit mass. If a contraction has a positive unit-preserving left inverse, the embedding is isometric. The same idea works on minimal tensor products.

For the SEP norm, the retraction gives both inequalities. A separable measurement on the large system restricts along the embedding to a separable measurement on the small system, while a separable measurement on the small system pulls back along the projection to a separable measurement on the large system. Hence embedded tensors have exactly the same SEP distinguishability norm.

Proof

We first record the elementary contraction fact. Let $T : V \to W$ be a positive map between GPTs (V, C, u) and (W, D, v), and assume $v \circ T = u$. If $x = x_+ - x_-$ with $x_\pm \in C$, then

$$Tx = Tx_+ - Tx_-, \qquad Tx_\pm \in D,$$

and

$$v(Tx_+) + v(Tx_-) = u(x_+) + u(x_-).$$

Taking the infimum over positive decompositions of x gives

$$\|Tx\|_W \leq \|x\|_V.$$

Apply this to the tensor maps

$$I = i_A \otimes i_B : V_{A_0} \otimes V_{B_0} \to V_A \otimes V_B, \qquad P = p_A \otimes p_B : V_A \otimes V_B \to V_{A_0} \otimes V_{B_0}.$$

Both maps are positive for the minimal tensor cones, because they send generators $a \otimes b$ with $a, b \geq 0$ to positive simple tensors. Both preserve the product order unit, and $PI = \mathrm{id}$. The contraction fact gives

$$\|IX\|_{A\otimes_{\min}B} \leq \|X\|_{A_0\otimes_{\min}B_0}$$

and, applying P,

$$\|X\|_{A_0\otimes_{\min}B_0} = \|PIX\|_{A_0\otimes_{\min}B_0} \leq \|IX\|_{A\otimes_{\min}B}.$$

Thus the minimal-composite base norm is preserved on the embedded copy.

It remains to compare the SEP distinguishability norms. Write a separable measurement on $A \otimes_{\min} B$ as a finite family $(E_j)_j$ with

$$E_j \in C_A^* \otimes_{\min} C_B^*, \qquad \sum_j E_j = u_A \otimes u_B.$$

Composing each effect with I gives

$$E_j \circ I \in C^*_{A_0} \otimes_{\min} C^*_{B_0}, \qquad \sum_j E_j \circ I = u_{A_0} \otimes u_{B_0}.$$

Hence $(E_j \circ I)_j$ is a separable measurement on $A_0 \otimes_{\min} B_0$, and therefore

$$\sum_j |E_j(IX)| = \sum_j |(E_j \circ I)(X)| \leq \|X\|_{\mathrm{SEP}(A_0,B_0)}.$$

Taking the supremum over all separable measurements on $A \otimes_{\min} B$ yields

$$\|IX\|_{\mathrm{SEP}(A,B)} \leq \|X\|_{\mathrm{SEP}(A_0,B_0)}.$$

Conversely, let $(F_j)_j$ be a separable measurement on $A_0 \otimes_{\min} B_0$. Composing with P gives

$$F_j \circ P \in C^*_A \otimes_{\min} C^*_B, \qquad \sum_j F_j \circ P = u_A \otimes u_B,$$

so $(F_j \circ P)_j$ is a separable measurement on $A \otimes_{\min} B$. Since $PI = \mathrm{id}$,

$$\sum_j |F_j(X)| = \sum_j |(F_j \circ P)(IX)| \leq \|IX\|_{\mathrm{SEP}(A,B)}.$$

Taking the supremum over all separable measurements on $A_0 \otimes_{\min} B_0$ gives the reverse inequality. Thus

$$\|IX\|_{\mathrm{SEP}(A,B)} = \|X\|_{\mathrm{SEP}(A_0,B_0)}.$$

For every nonzero X, the embedded tensor IX therefore has the same ratio

$$\frac{\|IX\|_{A\otimes_{\min} B}}{\|IX\|_{\mathrm{SEP}(A,B)}} = \frac{\|X\|_{A_0\otimes_{\min} B_0}}{\|X\|_{\mathrm{SEP}(A_0,B_0)}}.$$

Maximizing over X in $V_{A_0} \otimes V_{B_0}$ proves the theorem.

Gbit Consequence

Corollary 1. *Let B contain the gbit G_2 as a local order-retract subtheory. If A is non-simplicial, then*

$$R_{A\otimes_{\min} B}(\mathrm{SEP}) > 1.$$

The same conclusion holds with A and B interchanged.

Proof. Apply the theorem with $A_0 = A$ and $B_0 = G_2$. The source paper records the Namioka–Phelps result that

$$R_{A\otimes_{\min} G_2}(\mathrm{SEP}) = 1 \quad \text{if and only if} \quad C_A \text{ is simplicial.}$$

Thus, if A is non-simplicial, $R_{A\otimes_{\min} G_2}(\mathrm{SEP}) > 1$. Monotonicity then gives

$$R_{A\otimes_{\min} B}(\mathrm{SEP}) \geq R_{A\otimes_{\min} G_2}(\mathrm{SEP}) > 1.$$

□

Limitations and Novelty Check

This is not a proof of the full simplicial-cone conjecture. It proves only that known positive instances of SEP data hiding persist when both local systems are enlarged through positive unit-preserving retracts. The most direct new subcase is the one where a local system contains the gbit as such a retract.

Cheap duplicate checks were run against the run registry, solution, attempt, and proof-gap indexes for arXiv:1703.03392 and the phrases "R(SEP)", "simplicial", "gbit", "data hiding", "Namioka–Phelps", and "order retract". No existing run packet for this statement was found. Local later-source checks found improved bounds for the quantum local-operation problem but not an answer to the finite GPT simplicial-cone conjecture. Web checks found a later C*-algebraic Namioka–Phelps/Barker theorem, but not this finite GPT order-retract monotonicity statement.

References

1. L. Lami, C. Palazuelos, and A. Winter, *Ultimate data hiding in quantum mechanics and beyond*, arXiv:1703.03392.

2. I. Namioka and R. R. Phelps, *Tensor product of compact convex sets*, Pacific J. Math. 31 (1969), 469–480.

3. M. Musat and M. Rordam, *Entanglement in C*-algebras: tensor products of state spaces*, arXiv:2512.10410.

Human Review Recommendation

Review as a likely valid partial reduction. The proof is formal, but the key points to verify are the exact order-unit convention for allowed local maps and the assertion that composing separable effects with the retract maps preserves the SEP measurement class. Even if accepted, the result should not be counted as a full solution of the source conjecture.